

Digital is Different?

Chapter 2 — Managing Innovation (7th Ed.)

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Learning Objectives

By the end of this chapter, you will be able to:

- **Evaluate** whether digital innovation represents a fundamental shift or just a new toolset.
- **Analyze** the "Three S's" of digital impact: Scale, Scope, and Speed.
- **Understand** the economics of digital (Zero Marginal Cost).
- **Apply** the concept of the "Long Tail" to new business models.
- **Distinguish** between Pipeline and Platform business models.

Is Digital New?

Digital technology has existed for decades (transistors, microchips). So why the current hype?

The Inflection Point

We have reached a critical mass where digital is no longer just a "sector" (IT) but a "layer" across all sectors.

Key Drivers:

- **Moore's Law:** Exponential growth in computing power.
- **Connectivity:** Ubiquitous internet (IoT, 5G).
- **Data:** The ability to process massive datasets (Big Data).

The Three S's of Digital Impact

Digital changes the physics of innovation in three ways:

1. SPEED

Velocity

Transactions happen in milliseconds. Product lifecycles are shortening.

2. SCOPE

Convergence

Digital blurs boundaries (e.g., Apple is now a bank, a watchmaker, and a TV studio).

3. SCALE

Reach

A startup can reach a global audience instantly without physical infrastructure.

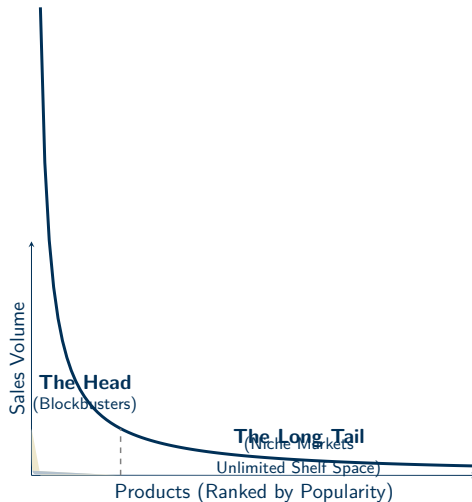
Traditional manufacturing follows "Diminishing Returns." Digital follows different rules.

- **Zero Marginal Cost:** It costs millions to develop software, but \$0.00 to replicate it.
- **Non-Rivalry:** My use of an app does not prevent you from using it (unlike a physical chair).
- **Network Effects:** The value of the service increases as more people use it (e.g., Facebook, Uber).

"Atoms are heavy and expensive; bits are weightless and free."

The Long Tail Effect

Digital markets allow us to sell niche products profitably.



Business Models: Pipeline vs. Platform

Digital shifts value from controlling the supply chain to orchestrating the connection.

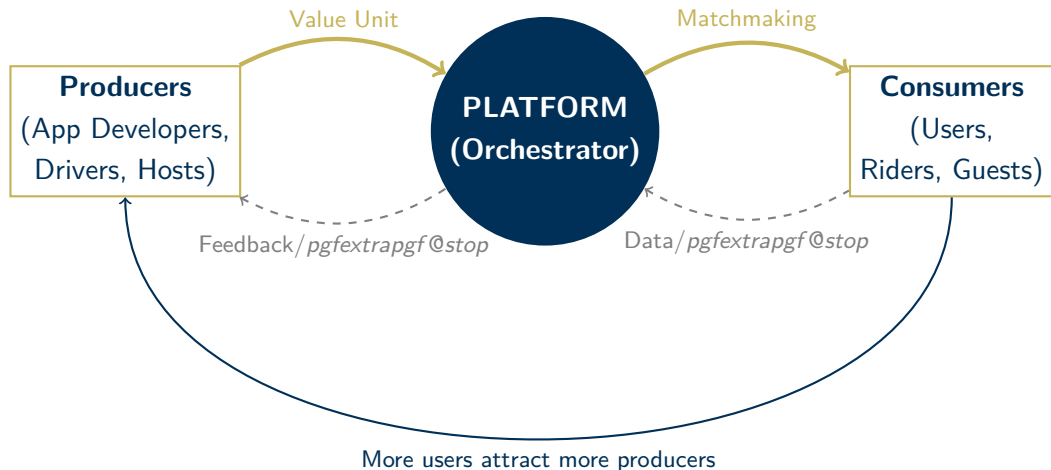
Pipeline (Traditional)

Linear flow: Design → Make → Sell.
Value is created inside the firm.

Platform (Digital)

Facilitates exchange between two groups (Producers & Consumers). Value is created *outside* the firm.

The Platform Ecosystem



Democratization: The Crowd Innovates

Digital lowers the barrier to entry for innovation.

- **User Innovation:** Users can now design, prototype (3D printing), and distribute their own solutions.
- **Crowdsourcing:** Companies can tap into the "cognitive surplus" of the global population (e.g., Wikipedia, Linux, LEGO Ideas).
- **Implication:** The smartest people don't work for you; you need to find a way to work with them.

The New Digital Toolkit

Digital provides new tools for the "Search, Select, Implement" process.

Innovation Phase	Digital Tool
Search	Big Data Analytics, Social Media Listening, AI Trend Spotting
Select	Crowdfunding (market validation), Prediction Markets
Implement	Rapid Prototyping, Digital Twins, Simulation, A/B Testing
Capture	Blockchain (IP protection), Freemium Models

Digital Darwinism

Technology evolves faster than organizational capacity to change.

The Dual Challenge (Ambidexterity):

1. **Digitize the Core:** Use digital to improve efficiency in the existing business (Exploitation).
2. **Digital Transformation:** Use digital to create entirely new business models (Exploration).

Many firms fail because they treat digital only as an efficiency tool, not a strategic shift.

Summary:

- Digital is different because of **Scale, Scope, and Speed**.
- It shifts economics from scarcity to abundance (Zero Marginal Cost).
- It favors **Platforms** over Pipelines and **Ecosystems** over solo firms.
- Leaders must embrace the **Long Tail** and open innovation to survive.

Thank You

Questions?